

Generative AI & Large Language Models (LLMs)

What is Generative AI?

Generative AI refers to AI systems that can generate new content such as text, images, audio, and code based on learned patterns from training data.

Large Language Models (LLMs)

Definition: LLMs are deep learning models trained on massive amounts of text data to understand and generate human language.

Capabilities:

- Text generation
- Question answering
- Translation
- Summarization
- Code generation

Popular LLMs

- OpenAI's GPT Series (GPT-3.5, GPT-4)
- Google's Bard/PaLM
- Meta's LLaMA
- Anthropic's Claude

Transformer Architecture

LLMs use Transformer architecture which includes:

- Self-Attention Mechanism
- Multi-Head Attention
- Positional Encoding
- Feed-Forward Networks

Applications

- Chatbots and Virtual Assistants
- Content Generation
- Code Assistance
- Research and Analysis
- Customer Support